Does trading gamification fuel bubbles?

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Abstract

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1 Introduction

> *For whatever reasons, markets now exhibit far more casino-like behavior than they did when I was young. The casino now resides in many homes and daily tempts the occupants.*
>
> — Warren Buffet (Annual Report for Berkshire Hathaway Inc., 2023)

Individual investors have quickly asserted themselves as a rising power in today's digital financial markets. In April 2025, retail traders accounted for 36% of the U.S. equity and options order flow — compared to less than 10% prior to the 2020 pandemic. In fact, retail volume now exceeds mutual funds and traditional hedge funds combined.[1] This shift extends beyond equities: Retail investors account for roughly 60% of same-day (0DTE) option volume (Bryzgalova et al., 2023), which itself represents more than half of total S&P options trading. Retail traders are also pivotal in cryptocurrency markets (Kogan et al., 2024).

Retail traders have fuelled the defining booms of recent years: the recurring meme-stock frenzies leading to 2000% jumps in GameStop and AMC share prices, the boom-bust cycles of cryptocurrencies, and the 2025 AI rally that produced the first $4 trillion firm, Nvidia. At the end of 2025, Cornelli et al. (2025) warned that retail exuberance—amplified by social interactions and fear of missing out—created conditions conducive to price bubbles in both equities and gold. Asness (2024) notes that markets have grown less efficient over the past three decades, and links this phenomenon to "*technology, gamified 24/7 trading on your phone, and social media in particular*" as the biggest culprits.

In this paper, we study how trading gamification affects price formation and market quality. Our central questions are whether gamified trading accelerates the emergence and collapse of asset bubbles, increases their magnitude, and widens the dispersion of outcomes across participants with different levels of financial literacy and trading experience.

To answer these questions, we design a controlled laboratory experiment following the canonical framework of Smith et al. (1988), in which participants trade a single asset over multiple periods

[1] See J.P. Morgan: Who is buying U.S. equities (June 4, 2025); Financial Times: Are retail traders the captains now? (December 9, 2025)

in a setting known to generate asset price bubbles. Our design varies two dimensions: interface type and market composition. First, following Chapkovski et al. (2026), participants trade in one of two interface conditions: (i) a non-gamified benchmark; (ii) a gamified environment with hedonic features such as confetti and achievement badges and informational features such as price-trend notifications. Second, we cross the gamification treatments with two market structures as in Farjam and Kirchkamp (2018): (a) human-only markets, and (b) hybrid markets where algorithmic traders may be present but participants cannot verify their presence. Crossing interface design with market composition yields four treatments. Since participants learn from experience in the Smith et al. (1988) framework, we employ a between-subjects design in which each participant is randomly assigned to a single treatment. We study how gamification impacts experience by repeating each market twice with the same group of participants.

MZ: [RESULTS PREVIEW HERE]

2 Literature review

Our paper contributes to a growing literature on the gamification of financial decision-making. In a series of online experiments, Chapkovski et al. (2025, 2026) show that gamified trading environments increase individual risk-taking, elevate trading volume, and lead to noisier and less disciplined strategies when participants trade against an exogenously generated price path. We instead focus on the *market-wide* consequences of gamification, particularly its effects on price formation and asset bubbles.

Empirical evidence on gamification in financial markets remains scarce, primarily due to data availability and identification challenges. Exploiting the staggered introduction of gamification features in U.S. brokerage platforms, Yelagin (2024) shows that gamification reduces retail traders' returns while benefiting market makers by making retail order flow less toxic. Arnold et al. (2022) find that push notifications alerting investors about large price swings nudge them into taking more leverage. Using proprietary data from a Czech gambling company, Hanousek et al. (2025) finds that exposure to gamification substantially prolongs gambling activity. Identification in these settings may be confounded by the endogenous adoption of gamification by platforms and traders. Our

controlled laboratory experiment avoids this concern by exogenously assigning gamification.

More broadly, we connect to a growing literature on trader behaviour in digital markets. Prior work shows that retail algorithmic trading does not mitigate bubbles (Asparouhova et al., 2024), that social communication tools induce pump-and-dump dynamics (Kluger and Saglam, 2025), and that retail traders misallocate resources toward financial data they struggle to process (**?**). Using transaction-level data, Kalda et al. (2021) find that investors take more risk on smartphones than on traditional platforms, while Barber et al. (2022) show that app design features such as Robinhood's *Top Movers* tab distort attention and reduce returns. We contribute by studying gamification as a salient design feature of trading apps and its market-wide effects.

Our paper also relates to a large literature on asset price bubbles. Early experimental work documents that bubbles arise even in environments with common knowledge of fundamentals (Smith et al., 1988). In field data, bubbles are linked to heterogeneous beliefs, extrapolation, overconfidence, and limits to arbitrage (Shiller, 2016; Barberis et al., 2015; Scheinkman and Xiong, 2003; Shleifer and Vishny, 1997). Using laboratory markets, Moinas and Pouget (2013) show that disagreement and short-sale constraints interact to fuel speculative bubbles, highlighting the role of belief dispersion in price dynamics. More recent work emphasizes the role of narratives and social interaction in amplifying speculative episodes (Shiller, 2019; Chinco, 2023). We add to this literature by studying how the trading user interface — by shaping attention and engagement — can influence the formation, magnitude, and collapse of asset price bubbles.

Finally, our paper connects to research in computer science, marketing, and psychology examining gamification and its influence on consumer behaviour. Deterding et al. (2011) define gamification as "the use of game design elements in non-game contexts." In this literature, gamification is often interpreted through the lens of hedonic consumption, whereby multisensory and affective elements of the user experience generate value beyond a product's utilitarian function (Hirschman and Holbrook, 1982; Heijden, 2004).

3 Experimental design

3.1 Trading game

Our experiment closely follows Smith et al. (1988, henceforth SSW). Subjects are randomly assigned to markets of six participants who trade a single asset in a limit order market over 15 periods. At the end of each period, the asset pays a stochastic dividend that is equally likely to be 0, 4, 8, or 20 experimental dollars (E$), yielding an expected dividend of E$8 per period. The asset's fundamental value therefore starts at E$120 and declines deterministically by E$8 each period, reaching zero at the end of the market.

Each period lasts one minute, following Farjam and Kirchkamp (2018) and Kopányi-Peuker and Weber (2020), so that each market lasts 15 minutes in total. Farjam and Kirchkamp (2018) argue that bubble magnitudes are nearly identical with one-minute periods and with the four-minute periods used in SSW. We therefore use shorter periods to allow each group to repeat the market twice within a session.

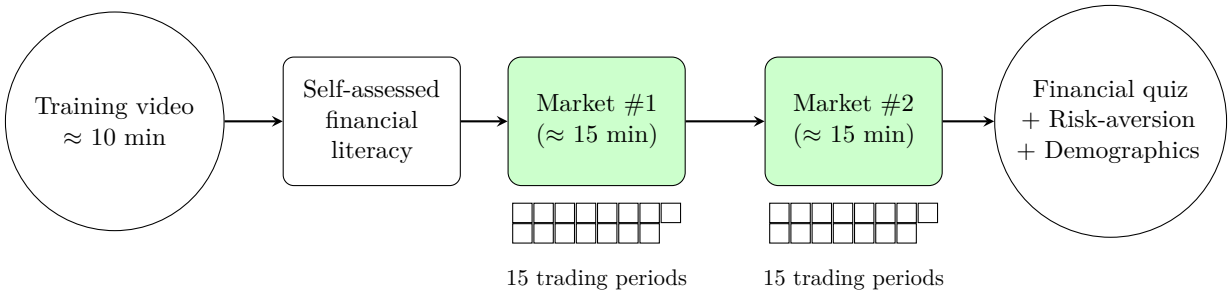
Half of the participants are endowed with 20 assets and E$2,600 in cash, while the other half receive 10 assets and E$3,800 in cash. At fundamental value, both endowments are worth E$5,000. We adopt a relatively high cash-to-asset ratio, which is known to amplify bubble formation (see, e.g., Caginalp et al., 1998), thereby increasing statistical power to detect treatment differences in bubble magnitude. Participants are not allowed to take negative asset holding (short selling) or borrow additional cash.

At the end of each period, after dividends are paid, we elicit participants' forecasts of the stock price in the subsequent period. Specifically, participants report a point forecast of the end-of-period price for the next period, along with their confidence in this assessment. Price forecasts are elicited using a continuous input field, while confidence is measured on a five-point Likert scale. Participants receive a bonus of E$100 if their forecast is within 5% of the realized price.

Finally, participants complete 13 financial literacy questions, reported in Appendix A. The questions are adapted from Fernandes et al. (2014) and include the three-item measure developed by Lusardi and Mitchell (2011), as well as two trading-specific questions based on the MiFID

questionnaire used in Inghelbrecht and Tedde (2024). To measure overconfidence and distinguish between subjective and objective financial literacy, participants also report their self-assessed financial knowledge on a scale from 0 (not at all knowledgeable) to 10 (very knowledgeable). This question follows Cupák et al. (2020) and is asked before the experiment begins to avoid contamination from trading outcomes or quiz difficulty. Figure 1 illustrates the experiment timing. Appendices B, C, and D contain the consent form, experimental instructions, and comprehension questions, respectively. Participants must correctly answer all six comprehension questions before trading begins.

Figure 1: **Experiment timing**



3.2 Treatments

The experimental design varies two key dimensions between participants. First, we manipulate the degree of gamification of the trading interface by introducing achievement badges and price-trend notifications intended to increase engagement and the salience of short-term price movements. Second, we vary participants' beliefs about market composition by introducing a mixed treatment in which traders may face algorithmic counterparts with positive probability. Crossing the two treatments allows us to study how gamification affects the emergence of bubbles both directly and through higher-order beliefs about other market participants.

The experiment is conducted in a computer lab at University of Calgary which can accommodate up to 36 participants per session. Each session consists of six independent markets with six participants per market. Participants are randomly allocated to markets subject to the session-level treatment quotas described in Table 1. Specifically, each session includes two gamified human-only markets, two non-gamified human-only markets, one gamified hybrid market, and one non-gamified

hybrid market.

Table 1: **Treatments for each experimental session**

Treatment cell	Market design	Traders	Markets (per session)	Participants (per session)
gh	Gamified	Human only	2	12
nh	Non-gamified	Human only	2	12
gm	Gamified	Mixed (human and algo)	1	6
nm	Non-gamified	Mixed (human and algo)	1	6
Total			**6**	**36**

We plan to run a total of ten laboratory sessions, yielding sixty markets and 360 participants overall.

3.2.1 Gamification strategies

The U.S. Securities and Exchange Commission (SEC) characterizes trading gamification elements as including "*social networking tools; games, streaks, contests with prizes; points, badges, and leaderboards; notifications; celebrations for trading; visual cues; curated ideas at order placement; subscriptions and membership tiers; and chatbots.*"[2] In this paper, we follow Chapkovski et al. (2026) and implement two gamification elements: achievement badges, representing hedonic gamification (defined in the spirit of Hirschman and Holbrook, 1982), as well as price trend notifications, representing informational gamification.

Achievement badges are awarded based on the number of trades executed by a participant within a market. Unlocking a badge triggers a visual animation consisting of falling confetti, a congratulatory message, and an animated GIF, as summarized in Table 2.

[2]See https://www.sec.gov/rules/other/2021/34-92766.pdf, pp. 7–8.

Table 2: **Achievement badges**

Badge	# trades	Message	GIF link
Bronze	10	Level up! Doing well 👍	https://bit.ly/3xtGrWk
Silver	25	You belong on the trading floor! 📈 💰	https://bit.ly/3l8kuqD
Gold	35	You are the money maker! 💰 💰	https://bit.ly/3nOwbog
Platinum	50	🚀 You are definitely going places! 🙌	https://bit.ly/3cNi6Be
Diamond	60	The Wolf of Wall Street 💎 🤲	https://bit.ly/3tMKDzE

Badge thresholds are calibrated using evidence from Farjam and Kirchkamp (2018), who report an average of 33.45 trades per participant in a 15-minute market. We therefore set the silver badge to be attainable for the average trader, while higher badge levels require above-average trading intensity.

A second gamification feature emphasizes price trends. Specifically, we introduce price-change alerts whenever the stock price increases or decreases for five consecutive trades. To facilitate clean identification, only one notification is displayed per price run. This ensures that trader responses can be attributed unambiguously to a five-trade run and avoids mechanically assigning greater empirical weight to longer runs.

We implement both gamification elements simultaneously. Because both features emphasize engagement and short-term price dynamics, we expect them to operate in the same direction. Combining them therefore increases statistical power by concentrating observations within a single gamified treatment cell. Figure 2 illustrates the design of the trading platform.

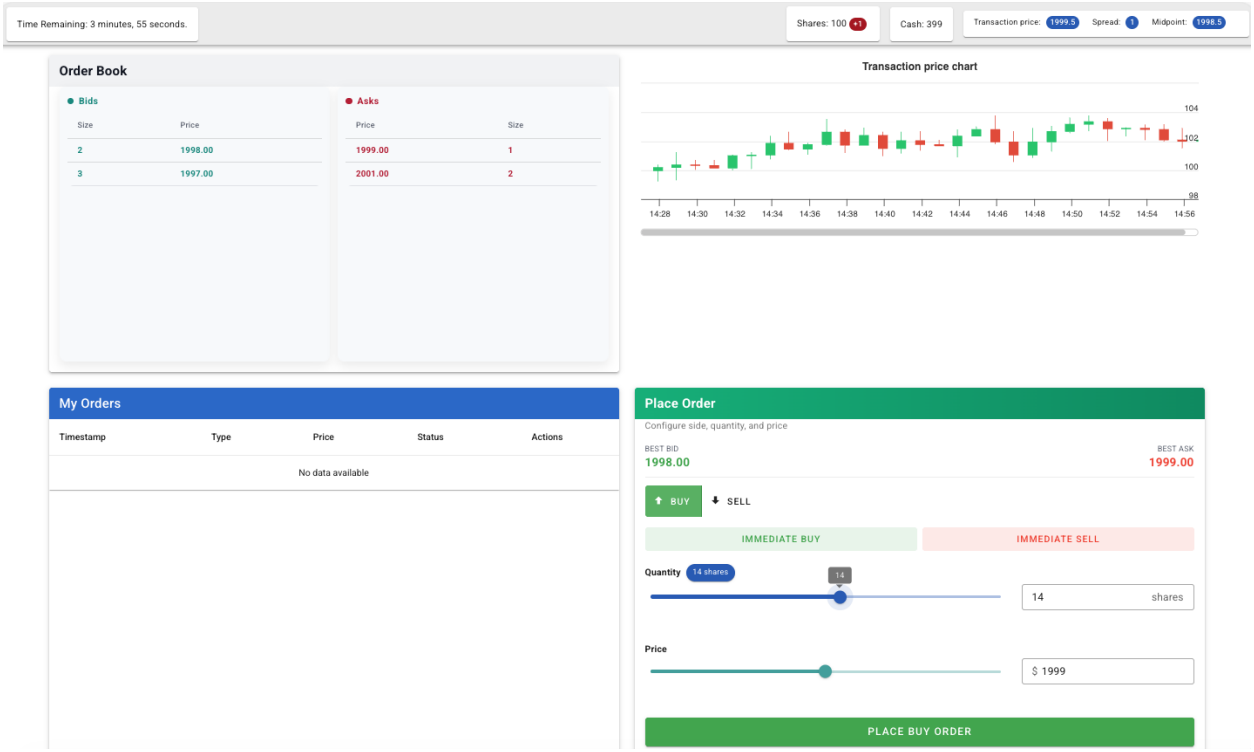


Figure 2: Trading platform user interface

3.2.2 Algorithmic traders

Following Farjam and Kirchkamp (2018), we implement a mixed treatment in which participants may trade against either human traders or algorithms. In this treatment, an algorithmic trader is introduced with probability 20%; otherwise, participants trade exclusively with humans. Participants are informed of the possibility of algorithmic trading but not of the exact probability that an algorithm is present. At the end of each market in the mixed treatment, participants are asked whether they believe algorithmic traders were present.

Our primary interest is not in markets in which participants actually trade against algorithms, but in how the possibility of algorithmic trading affects beliefs and behaviour. Even in markets that consist entirely of human traders, participants in the hybrid treatment may perceive the average trader as less susceptible to gamification and therefore rationally refrain from trend-chasing in anticipation of others' behaviour.

When present, the algorithmic trader submits between zero and three orders per period with

equal probability, randomly choosing between market and limit buy and sell orders. Limit orders are placed sufficiently far from the best bid and ask to avoid systematically affecting prices. The algorithmic trader's strategy is not disclosed to participants, reflecting the limited and imperfect information human traders have about algorithmic trading in real-world markets, where such strategies are proprietary and opaque (as argued in Farjam and Kirchkamp, 2018).

4 Hypotheses development

Hypothesis 1 (Bubble size). *Mispricing is larger in gamified treatments.*

We conjecture that gamification amplifies asset price bubbles, in that prices in gamified markets will deviate further from fundamental value. The rationale is that gamification increases hedonic utility from trading itself (Hirschman and Holbrook, 1982), generating excitement that amplifies bubbles (Andrade et al., 2016). Further, we hypothesize that price trend notifications amplify feedback trading, which Haruvy and Noussair (2006) find to be a key driver of asset bubbles.

Following Asparouhova et al. (2024), we compute average and absolute mispricing for market m and period k:

$$\text{AverageMispricing}_{k,m} = \frac{\sum_{t \in \mathcal{T}_{k,m}} (p_t - v_k)}{|\mathcal{T}_{k,m}|}, \quad \text{AbsoluteMispricing}_{k,m} = \frac{\sum_{t \in \mathcal{T}_{k,m}} |p_t - v_k|}{|\mathcal{T}_{k,m}|}, \tag{1}$$

where $\mathcal{T}_{k,m}$ is the set of trades in period k of market m (indexed by t), p_t is the transaction price for trade t, and v_p is the fundamental value in period p. We further compute the absolute mispricing ratio (Palan, 2013), defined as absolute mispricing normalized by the fundamental value:

$$\text{AbsoluteMispricingRatio}_{k,m} = \frac{\text{AbsoluteMispricing}_{k,m}}{v_k}. \tag{2}$$

Hypothesis 2 (Volatility). *The number of price surges and crashes is higher in gamified treatments.*

Chapkovski et al. (2026) show that gamification leads to higher trading volume: Hedonic features make trading intrinsically rewarding, while notifications increase salience and nudge participants into action. Haruvy and Noussair (2006) identify feedback traders, that is trend-chaser investors

who buy as prices rise and sell as they fall, as central to bubble dynamics. Feedback traders are less sophisticated than either speculators (who trade on expectations) or passive traders (who anchor to fundamentals), and therefore most susceptible to gamification. Therefore, we conjecture that trading gamification accelerates both bubble formation and collapse by disproportionately increasing feedback trading.

As in Asparouhova et al. (2024), we define price surges and crashes as periods in which the asset return deviates by more than two standard deviations from the average return across all markets in the same repetition (first or second).

To capture longer-run bubble dynamics, we define bubble runs based on normalized mispricing (Noussair et al., 2001). Let $p_{k,m}$ denote the average transaction price in period k of market m, and let v_k denote the fundamental value. A period is classified as a bubble period if $(p_{k,m} - v_k)/v_k$ exceeds two standard deviations above the mean mispricing across all markets within the same repetition. A bubble run is then defined as a maximal sequence of consecutive bubble periods. We use the number of bubble runs per market as our outcome of interest.

Hypothesis 3 (**Wealth inequality**). *Gamification increases wealth inequality.*

Financially literate traders are better equipped to resist behavioural biases (Lusardi and Mitchell, 2011). At the same time, Chapkovski et al. (2025) find that less sophisticated traders are more susceptible to gamification. In a market setting, we conjecture that the mistakes of inexperienced investors create opportunities for sophisticated traders to exploit. Gamification therefore transfers wealth from less to more sophisticated participants, increasing inequality. We predict higher dispersion in final wealth in gamified markets, as measured by the standard deviation of payoffs or the Gini (1936) coefficient

$$G_m = \frac{\sum_i^n \sum_j^n |\pi_{i,m} - \pi_{j,m}|}{2n^2 \mathbb{E}\pi}, \tag{3}$$

where $\pi_{i,m}$ stands for the payoff of trader i in market m.

Hypothesis 4 (**Experience**). *Experience reduces bubbles less in gamified than in non-gamified markets.*

Smith et al. (1988) and Dufwenberg et al. (2005) show that repeated participation in asset

markets reduces the likelihood and magnitude of bubbles. Gamification may weaken the corrective role of experience by redirecting participants' attention toward price dynamics and hedonic interface elements rather than fundamentals.

We therefore predict that experience attenuates bubbles to a lesser extent in gamified markets than in control markets.

Hypothesis 5 (**Feedback trading**). *Gamification amplifies feedback trading and dampens fundamental value trading.*

We classify traders as in Haruvy and Noussair (2006) by comparing position changes to (ex dividend) price movements. First, a trader is classified as *feedback* if the change in holdings for period k has the same sign as the price change from $k-2$ to $k-1$ (that is, trend chasing). Second, a trader is a *speculator* if the holding change in k has the same sign as the subsequent price change from k to $k+1$ (i.e., the speculator anticipates price changes). Finally, a trader is *fundamental* if the change in holdings has the opposite sign as the deviation of price from fundamental value, that is the participant trades toward fundamentals. Each trader accumulates a score for each type, incremented by one for each period consistent with that type. We classify traders by their highest score, provided it reaches 8 (a majority of the 15 periods). Traders with no score above 8 are classified as "other." Ties are split equally: 50% weight to each type for two-way ties, one-third for three-way ties. We conjecture that in gamified treatments there are more feedback traders and fewer fundamental traders.

Hypothesis 6 (**Algo trading**). *The impact of gamification on market outcomes is lower when traders believe algorithms may be present.*

Following Farjam and Kirchkamp (2018), we compare markets where participants know they face only humans to markets where algorithms may be present but are not. Gamification may operate partly through beliefs about others: if participants expect other traders to chase trends on gamified platforms, it becomes rational to maintain the bubble. Algorithmic traders, in contrast, are immune to the behavioural cues of a gamified interface. We therefore predict that the effects of gamification on mispricing, the frequency of price surges and crashes, and wealth inequality are

attenuated when participants anticipate that not all traders are human.

Hypothesis 7 (**Beliefs**). *Gamification strengthens trend extrapolation in beliefs, but the effect is attenuated when algorithmic traders may be present.*

Haruvy et al. (2007) find that individual's beliefs about prices are adaptive and depend on past trends. We postulate that gamification reinforces this tendency by highlighting trends and rewarding action.

We measure trend extrapolation as the sensitivity of price forecasts to recent price changes. Let $F_{i,k+1}$ denote subject i's forecast of the end-of-period price in period $k+1$, elicited at the end of period k, and let P_k denote the realized end-of-period price in period k. We estimate panel regressions of $F_{i,k+1} - P_k$ on the recent price change $\Delta P_k = P_k - P_{k-1}$ and its interactions with treatment indicators:

$$\begin{aligned} F_{i,k+1} - P_k = \alpha + \beta_0\, \Delta P_k + \beta_1\, \text{Gamified}_m + \beta_2\, \text{Hybrid}_m + \\ + \Delta P_k\, (\gamma_0 \text{Gamified}_m + \gamma_1 \text{Hybrid}_m + \gamma_2 \times \text{Gamified}_m \times \text{Hybrid}_m) + \\ + \text{Controls} + \varepsilon_{i,k}. \end{aligned} \tag{4}$$

In particular, we test whether the sensitivity of forecasts to recent price trends increases under gamification ($\gamma_0 > 0$) and whether this increase is attenuated in the hybrid treatment ($\gamma_2 < 0$).

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Online Appendix

January 21, 2026

A Financial literacy quiz

1. Suppose you had $100 in a savings account and the interest rate was 3% per year. After 4 years, how much do you think you would have in the account if you left the money to grow?
   (a) More than $112
   (b) Exactly $112
   (c) Less than $112
   (d) Don't know / Not sure
   (e) Prefer not to say
2. Imagine that the interest rate on your savings account was 1% per year and inflation was 2% per year. After 1 year, with the money in this account, would you be able to buy
   (a) More than today
   (b) Exactly the same as today
   (c) Less than today
   (d) Don't know / Not sure
   (e) Prefer not to say
3. When placing an order, you may see several prices: the last traded price, the ask price, the bid price, and the closing price. What is the bid price?
   (a) The highest price a buyer is currently willing to pay.
   (b) The price you personally choose when submitting a buy order.
   (c) The lowest price a seller is currently willing to accept.
   (d) The most recent price at which the security traded.
   (e) Don't know / Not sure
   (f) Prefer not to say
4. A 15-year mortgage typically requires higher monthly payments than a 30-year mortgage, but the total interest paid over the life of the loan will be less.
   (a) True
   (b) False
   (c) Don't know / Not sure
   (d) Prefer not to say
5. Normally, which asset described below displays the highest fluctuations over time?
   (a) Savings accounts
   (b) Stocks
   (c) Bonds

(d) Don't know / Not sure

(e) Prefer not to say

6. When an investor spreads his money among different assets, does the risk of losing a lot of money:

   (a) Increase

   (b) Decrease

   (c) Stay the same

   (d) Don't know / Not sure

   (e) Prefer not to say

7. Considering a long time period (for example, 10 or 20 years), which asset described below normally gives the highest return?

   (a) Savings accounts

   (b) Stocks

   (c) Bonds

   (d) Don't know / Not sure

   (e) Prefer not to say

8. Do you think that the following statement is true or false? "If you were to invest $1,000 in a stock mutual fund, it would be possible to have less than $1,000 when you withdraw your money."

   (a) True

   (b) False

   (c) Don't know / Not sure

   (d) Prefer not to say

9. When viewing a security's quote, you may see both a bid price and an ask price. What is the bid–ask spread?

   (a) The difference between the lowest ask price and the highest bid price.

   (b) The average of the bid price and the ask price.

   (c) The fee the broker charges each time you trade.

   (d) The change in the stock price from market open to market close.

   (e) Don't know / Not sure

   (f) Prefer not to say

10. Which of the following statements is correct?

   (a) Once one invests in a mutual fund, one cannot withdraw the money in the first year

   (b) Mutual funds can invest in several assets, for example invest in both stocks and bonds

   (c) Mutual funds pay a guaranteed rate of return which depends on their past performance

(d) None of the above

(e) Don't know / Not sure

(f) Prefer not to say

11. Which of the following statements is correct? If somebody buys a bond of firm B:

    (a) She owns a part of firm B

    (b) She has lent money to firm B

    (c) She is liable for the company's debts

    (d) None of the above

    (e) Don't know / Not sure

    (f) Prefer not to say

12. Suppose you owe $3,000 on your credit card. You pay a minimum payment of $30 each month. At an annual percentage rate of 12% (or 1% per month), how many years would it take to eliminate your credit card debt if you made no additional new charges?

    (a) Less than 5 years

    (b) Between 5 and 10 years

    (c) Between 10 and 15 years

    (d) Never

    (e) None of the above

    (f) Don't know / Not sure

    (g) Prefer not to say

B Consent form

Dear Participant,

You are invited to take part in an economic experiment studying trading behaviour in financial markets. This project is funded by the Social Sciences and Humanities Research Council of Canada. You will be asked to trade stocks on a virtual financial market with other participants.

Your participation in this study is entirely voluntary and presents no known risks or burdens. You will receive a fixed payment for participating, as well as additional compensation based on your trading performance and your answers to a financial literacy quiz. You can withdraw from the study at any time without penalty. However, if you withdraw before the end of the experiment, you will receive only a partial payment reflecting the time spent.

All data generated during this study will remain confidential. You will be asked to provide some basic demographic information, but none of this information will be used to personally identify you and no personal information will appear in any published study. Your decision to participate will remain completely confidential and only the researchers will have access to individual data. We do not collect any data that can personally identify you.

Data will be stored electronically on a secure server accessible only by the research team using a secure password. Data will be retained in de-identifiable format for 7 years and discarded afterwards, in accordance with standard practices of the American Psychological Association. As required by the guidelines of the American Economic Association, during the publication process the data may be shared with other investigators for re-analysis. However, only details sufficient to permit replication will be shared, ensuring security and confidentiality.

This study has been reviewed by the Research Ethics Board at the University of Calgary. The research study you are participating in may be reviewed for quality assurance to ensure that required laws and guidelines are followed. If chosen, representative(s) of the Human Research Ethics Program (HREP) may access study-related data and/or consent materials as part of the review. All information accessed by the HREP will be upheld to the same level of confidentiality stated by the research team.

If you are interested, you can receive a summary of the research study by emailing the primary researchers.

If you wish to obtain further information regarding your rights as a participant, you may contact the Conjoint Faculties Research Ethics Board (CFREB) at the University of Calgary at `cfreb@ucalgary.ca` or (403) 220-8640.

By proceeding, I agree to participate in research. I understand that I can withdraw my consent to participate at any time and that by giving consent I am not giving up any of my legal rights.

I have read and understood the information above.

C Instructions

Welcome to this trading experiment! Please read these instructions carefully. Your payment will be based on your performance in the trading game and your answers to a financial literacy quiz.

Overview

- You will trade a single stock in a market with five other participants. The experiment consists of two consecutive markets with the same group of traders.
- At the beginning of each market you will receive an endowment of either 20 shares and E$2,600 in cash, or 10 shares and E$3,800 in cash.
- Each market consists of 15 trading periods, each lasting 1 minute (15 minutes total per market). At the end of each period, you will be asked to forecast the price at the end of the following period.
- Finally, at the end of both markets you will be asked to complete a financial literacy quiz.

Algorithmic traders

- Be aware that algorithmic traders may be operating in your market, alongside human participants.
- You will not be informed whether any algorithms are present. If present, they submit orders automatically according to a pre-programmed strategy that is not disclosed to participants.
- At the end of the experiment, you will be asked whether you believe algorithmic traders were present.

The Stock

- The stock pays a dividend at the end of every period. The dividend is equally likely to be E$0, E$4, E$8, or E$20, yielding an expected dividend of E$8 per period. Dividends are drawn independently each period.
- Because the stock pays dividends only during the 15 periods of the market, its fundamental value declines over time. The fundamental value equals E$8 multiplied by the number of remaining periods (including the current period). It starts at E$120 in period 1 and declines by E$8 each period, regardless of the actual dividend realized. By period 15, the fundamental value is E$8. After the final dividend is paid, the stock is worthless.

Trading Rules You will trade using a limit order book. You may submit buy or sell orders at prices you specify, or you may trade immediately against existing orders in the book.

- You may only buy shares if you have sufficient cash to pay for them.
- You may only sell shares you own. Short selling is not permitted.
- You cannot borrow additional cash.

Price Forecasting At the end of each period, after dividends are paid, you will forecast the end-of-period stock price for the next period. You will also report your confidence in this forecast. If your forecast is within 5% of the realized price, you earn a bonus of E$100.

Payment

- Your final payoff equals your cash holdings at the end of the second market, including any period 15 dividends, plus any forecast bonuses earned across both markets.
- Your earnings will be converted from experimental dollars (E$) to real money at a rate of [TBD].
- You will earn [TBD] for each correct answer in the financial literacy quiz.

Important Notes

- Your trading screen will show the current order book, recent transaction prices, your cash balance, and your share holdings.
- Each period lasts exactly 1 minute. A timer will indicate the time remaining.
- After completing the first market, you will trade a second market with the same group of participants. The second market is independent: you will receive a fresh endowment, and the stock's fundamental value resets to E$120.

Training Before the main markets begin, you will watch a short video explaining the trading interface and complete a comprehension quiz. You must answer all quiz questions correctly before proceeding.

D Comprehension quiz

1. At the end of each period, the asset pays a dividend. Which of the following statements is correct?
   (a) The dividend is always E$8.
   (b) The dividend is equally likely to be E$0, E$4, E$8, or E$20.
   (c) The dividend depends on the asset's trading price.
   (d) The dividend is determined by your trading activity.
2. At the end of each period, you will forecast the stock price for the next period. When do you earn a bonus?
   (a) If your forecast is within 5% of the realized price.
   (b) If your forecast exactly equals the realized price.
   (c) You earn E$100 for every forecast submitted.
   (d) If your forecast is within 1% of the realized price.
3. Suppose the dividend in period 3 is E$20. How does this affect the fundamental value in period 4?
   (a) The fundamental value increases by E$20.
   (b) The fundamental value is unaffected by the realized dividend.
   (c) The fundamental value decreases by E$20.
   (d) The fundamental value equals E$20.
4. Which of the following best describes how your final payoff is determined?
   (a) Your cash holdings at the end of period 15, including period 15 dividends.
   (b) The number of shares you hold at the end of period 15.
   (c) Your cash holdings, including period 15 dividends, plus any forecast bonuses earned.
   (d) Your cash holdings at the start of period 15, plus any forecast bonuses earned.
5. Which of the following actions are **not** permitted in this market?
   (a) Selling shares you do not own (short selling).
   (b) Borrowing cash to purchase additional shares.
   (c) Buying shares only if you have sufficient cash.
   (d) Both (a) and (c).